

HASAN BALCI

PERSONAL INFORMATION

Date of birth: 27 March 1989
Nationality: Turkish
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EDUCATION

PhD in Computer Engineering 05/09/2016 - 18/08/2022
Bilkent University, Turkey
Supervisor: Prof. Dr. Ugur Dogrusoz
Thesis Title: "Fast Compound Graph Layout with Constraint Support"
CGPA: 3.77/4.00

MS in Computer Engineering 06/09/2012 - 05/08/2015
Bilkent University, Turkey
Supervisor: Prof. Dr. Ugur Gudukbay
Thesis Title: "Sun Position Estimation on Time-lapse Videos for Augmented Reality Applications"
CGPA: 3.69/4.00

BS in Computer Engineering 27/08/2007 - 17/06/2012
Bilkent University, Turkey
CGPA: 3.58/4.00

RESEARCH INTERESTS

My research interests lie at the intersection of graph visualization, graph algorithms, and computational biology. I develop computational and interactive methods for representing, analyzing, and visualizing complex networks, with a particular focus on biological networks and pathway maps. I combine graph-based methods with AI-assisted and human-in-the-loop approaches to support biological knowledge curation and interpretation. I am also interested in knowledge graphs and graph-based approaches to scientific data integration and analysis.

RESEARCH EXPERIENCE

Postdoctoral Researcher 03/06/2024 - 02/06/2026
Computational Biology Branch, National Library of Medicine (NLM), NIH, USA
Supervised by Dr. Augustin Luna

- Designed and implemented SBGNFlow, an AI-assisted and interactive workflow for digitizing hand-drawn SBGN maps, merging/splitting pathway maps, and performing user-guided layout refinements [\[16\]](#) [\[17\]](#)
- Worked on updating the SBGN specifications, including refining notation rules, ensuring consistency across standards, and incorporating community feedback [\[4\]](#) [\[14\]](#)

Postdoctoral Researcher 30/01/2023 - 29/01/2024
Maastricht Center for Systems Biology (MaCSBio), Maastricht University, the Netherlands
Supervised by Dr. Martina Summer-Kutmon

- Worked on a project that leverages knowledge graphs on drug repurposing for COVID-19 and its long-term effects (post-COVID)

- Worked on a project that explores the innate immune response against live and inactivated SARS-CoV-2 virus [\[15\]](#)
- Contributed to WikiPathways biological pathway platform [\[6\]](#)

Postdoctoral Researcher

01/09/2022 - 25/01/2023

i-Vis Research Lab, Bilkent University, Turkey

Supervised by Dr. Ugur Dogrusoz

- Worked on design and implementation of a research project named “Effective Analysis of Big Data Through Graph Visualization with A Unified Complexity Management Framework” supported by The Scientific and Technological Research Council of Turkey (TÜBİTAK) [\[1\]](#) [\[2\]](#)
- Contributed to the design and implementation of SyBValS, a web service for validation and error resolution of biological pathway maps [\[3\]](#)

Research Assistant

05/09/2016 - 18/08/2022

i-Vis Research Lab, Bilkent University, Turkey

Supervised by Dr. Ugur Dogrusoz

- Designed and developed SyBLaRS, a web service for biological pathway layout and rendering [\[7\]](#)
- Designed and developed fCoSE (fast Compound Spring Embedder) [\[8\]](#), a compound graph layout algorithm distributed as cytoscape-fcose and used in AWS Workload Discovery, AWS Graph Explorer, and Mermaid — 6M+ weekly npm downloads and 178 GitHub stars.
- Contributed to the design and development of CoSEP (Compound Spring Embedder with Ports) [\[9\]](#), a layout algorithm for compound graphs with port constraints.
- Contributed to the design and implementation of a research project named “Efficient Layout Algorithms for Compound Graphs With Support for Ports and Constraints” supported by TÜBİTAK
- Participated in Google Summer of Code 2017 with a project which aims visual quality and performance improvements in CoSE (Compound Spring Embedder) layout algorithm
- Led development and maintenance of Newt Pathway Viewer and Editor which is a web-based, open source viewer and editor for pathways in SBGN, SBML and SIF formats [\[10\]](#)
- Contributed to the development of Cytoscape.js extensions for compound graph complexity management, including cytoscape-expand-collapse and cytoscape-view-utilities [\[11\]](#), used in AWS Workload Discovery and AWS Graph Explorer — 8,600+ weekly npm downloads and 140 GitHub stars for cytoscape-expand-collapse.

Research Assistant

06/09/2012 - 05/08/2015

Mod-Vis Research Group, Bilkent University, Turkey

Supervised by Dr. Ugur Gudukbay

- Contributed to the research project named “An Augmented Reality Environment for Interactive Crowd Simulation” supported by TÜBİTAK and developed a method for sun position estimation and tracing in time-lapse videos [\[12\]](#)

Undergraduate Research

09/2011 - 06/2012

Bilkent University, Turkey

- Worked on senior design project named “Turkish Sign Language Converter via Microsoft Kinect” that converts the signs in Turkish Sign Language to textual form

PUBLICATIONS

- [1] U. Dogrusoz, H. Islam, and **H. Balci**, "HySE: a force-directed layout algorithm for directed and mixed graphs", *Computer Graphics Forum*, e70586, 2026. DOI: [10.1111/cgf.70586](https://doi.org/10.1111/cgf.70586)
- [2] O. Zafar, U. Dogrusoz, **H. Balci**, and A. F. Halac, "CMGV: Algorithms and a unified framework for complexity management in graph visualization", *Information Visualization*, 25(2), pp. 192-208, 2026. DOI: [10.1177/14738716251383173](https://doi.org/10.1177/14738716251383173)
- [3] Y. Z. Ozgul, U. Dogrusoz, and **H. Balci**, "SyBValS: a validation and error resolution service for biological pathway maps", *BMC Genomics*, 27(106), 2026. DOI: [10.1186/s12864-025-12454-4](https://doi.org/10.1186/s12864-025-12454-4)
- [4] **H. Balci**, A. Rougny, R. Overall, I. Balaur, ..., and A. Luna, "Systems biology graphical notation: process description language level 1 version 2.1", *Journal of Integrative Bioinformatics*, 20250018, 2026. DOI: [10.1515/jib-2025-0018](https://doi.org/10.1515/jib-2025-0018)
- [5] A. Niarakis, G. An, L. Ladeira, ..., **H. Balci**, ..., and R. Laubenbacher, "Building Immune Digital Twins: An International and Transdisciplinary Community Effort", *ImmunoInformatics*, 100060, 2025. DOI: [10.1016/j.immuno.2025.100060](https://doi.org/10.1016/j.immuno.2025.100060)
- [6] A. Niarakis, R. Laubenbacher, G. An, Y. Ilan, ..., **H. Balci**, ..., and J. A. Glazier, "Immune digital twins for complex human pathologies: applications, limitations, and challenges", *NPJ Systems Biology and Applications*, 10(01), 141, 2024. DOI: [10.1038/s41540-024-00450-5](https://doi.org/10.1038/s41540-024-00450-5)
- [7] A. Agrawal, **H. Balci**, K. Hanspers, S. L. Coort, M. Martens, D. N. Slenter, ..., and A. R. Pico, "WikiPathways 2024: next generation pathway database", *Nucleic Acids Research*, 52(D1), pp. D679-D689, 2024. DOI: [10.1093/nar/gkad960](https://doi.org/10.1093/nar/gkad960)
- [8] **H. Balci**, U. Dogrusoz, Y. Z. Ozgul, and P. Atayev, "SyBLaRS: A web service for laying out, rendering and mining biological maps in SBGN, SBML and more", *PLOS Computational Biology*, 18(11), pp. 1-12, 2022. DOI: [10.1371/journal.pcbi.1010635](https://doi.org/10.1371/journal.pcbi.1010635)
- [9] **H. Balci** and U. Dogrusoz, "fCoSE: a fast compound graph layout algorithm with constraint support", *IEEE Transactions on Visualization and Computer Graphics*, 28(12), pp. 4582-4593, 2022. DOI: [10.1109/TVCG.2021.3095303](https://doi.org/10.1109/TVCG.2021.3095303)
- [10] A. Okka, U. Dogrusoz, and **H. Balci**, "CoSEP: a compound spring embedder layout algorithm with support for ports", *Information Visualization*, 20(2-3), pp. 151-169, 2021. DOI: [10.1177/14738716211028136](https://doi.org/10.1177/14738716211028136)
- [11] **H. Balci**, M. C. Siper, N. Saleh, I. Safarli, L. Roy, M. Kilicarslan, R. Ozaydin, A. Mazein, C. Auffray, O. Babur, E. Demir, and U. Dogrusoz, "Newt: a comprehensive web-based tool for viewing, constructing, and analyzing biological maps", *Bioinformatics*, 37(10), pp. 1475-1477, 2021. DOI: [10.1093/bioinformatics/btaa850](https://doi.org/10.1093/bioinformatics/btaa850)
- [12] U. Dogrusoz, A. Karacelik, I. Safarli, **H. Balci**, L. Dervishi, and M. C. Siper, "Efficient methods and readily customizable libraries for managing complexity of large networks", *PLOS ONE*, 13(5): e0197238, 2018. DOI: [10.1371/journal.pone.0197238](https://doi.org/10.1371/journal.pone.0197238)
- [13] **H. Balci** and U. Gudukbay, "Sun position estimation and tracking for virtual object placement in time-lapse videos", *Signal, Image and Video Processing*, 11(5), pp. 817-824, 2017. DOI: [10.1007/s11760-016-1027-x](https://doi.org/10.1007/s11760-016-1027-x)
- [14] **H. Balci**, A. Rougny, L. Ladeira, H. Mi, ..., and A. Luna, "Systems biology graphical notation: activity flow language level 1 version 1.3", *Journal of Integrative Bioinformatics*, 2026. (Accepted/In press)
- [15] R. J. Jacobi, J. Cvetkovic, **H. Balci**, D. Heylen, A. Miranda-Bedate, E. Pinelli, M. Kutmon, and M. D.B. van de Garde, "SARS-CoV-2 induces robust antiviral innate immunity in PBMCs that is partially mirrored by inactivated virus". *Access Microbiology*, 001299-v1, 2026. (Preprint)

- [16] **H. Balci** and A. Luna, "SketchLay: User-Guided Force-Directed Graph Layout through Freehand Sketching". Invited major revision at *IEEE Transactions on Visualization and Computer Graphics*. Shorter version: "User-Guided Force-Directed Graph Layout", arXiv preprint arXiv:2506.15860, 2025. DOI: [10.48550/arXiv.2506.15860](https://doi.org/10.48550/arXiv.2506.15860)
- [17] **H. Balci** and A. Luna, "SBGNFlow: An AI-Assisted & Interactive Workflow for Generation, Merging & Layout of Pathway Maps". (In preparation)

CONFERENCES / WORKSHOPS / MEETINGS / TALKS

ISMB Conference 2026 - BioVis COSI

12/07/2026 - 16/07/2026

Washington, DC, USA

Accepted for a talk titled "SBGNFlow: An AI-Assisted & Interactive Workflow for Generation, Merging & Layout of Pathway Maps".

COMBINE Workshop 2025

20/10/2025 - 23/10/2025

Participated online

Hosted a breakout session on SBGN, collaboratively working on the update of the Activity Flow and Hybrid PD-AF specifications.

Gave a lightning talk titled "From Sketch to SBGN: An AI-Assisted Workflow for Map Creation".

Disease Maps Community Online Meetings

30/01/2025

Participated online

Gave a talk titled "SBGN Hybrid PD-AF Notation".

NIH Artificial Intelligence Symposium 2025

16/05/2025

NIH Bethesda Campus, MD, USA

Poster: Automatic Conversion of Hand-Drawn to Interactive Pathway Diagrams Using Large Language Models

HARMONY Workshop 2025

15/04/2025 - 18/04/2025

Participated online

Hosted a breakout session on SBGN, collaboratively working on the update of the Activity Flow specification.

Disease Maps Community Online Meetings

30/01/2025

Participated online

Gave a talk titled "Advancing SBGN: Insights into Layout and AI-Assisted Tools".

COMBINE Workshop 2024

02/09/2024 - 05/09/2024

Participated online

Hosted a breakout session on SBGN, collaboratively working on the update of the Process Description specification.

BioHackathon Europe

30/10/2023 - 03/11/2023

Participated online

Worked on a project named "Extending interoperability of experimental data using modular queries across biomedical resources".

Y. Gadiya, A. Ammar, E. Willighagen, D. Martinat, A. C. Sima, **H. Balci** & T. Abbassi-Daloi, (2023). BioHackEU23 report: Extending interoperability of experimental data using modular queries across biomedical resources, 2023. DOI: [10.37044/osf.io/mhsqp](https://doi.org/10.37044/osf.io/mhsqp)

Building Immune Digital Twins Workshop

15/05/2023 - 02/06/2023

Paris, France

Selected to join a three-week workshop that brought together researchers across disciplines for activities ranging from extended active teamwork on specific immune digital twin projects to lectures, discussion and working groups, and brainstorming sessions

Bioinformatics & Systems Biology Conference 2023

09-10/05/2023

Egmond aan Zee, The Netherlands

Poster: From WikiPathways to Drug Repurposing for COVID-19: Utilizing Knowledge Graphs to Identify Novel Therapies

Demo: SyBLaRS: Systems Biology Layout and Rendering Service

7th Disease Maps Community Meeting

03-05/04/2023

Maastricht, The Netherlands

Member of Local Organization Committee

PROFESSIONAL SERVICE

Editor

- Systems Biology Graphical Notation (SBGN) [\[1\]](#) [\[13\]](#) 01/2025 - present

Scientific Communities

- Building Immune Digital Twins (BIDT) Community [\[4\]](#) [\[5\]](#) 05/2023 - present
- Disease Maps Community 02/2023 - present
- WikiPathways Community 02/2023 - present

Reviewer

- ISMB NetBio COSI 2024 - present
- Various conferences and journals including EuroVis and PLOS One 2025 - present

Additional Professional Service

- Judge, NIH 2026 Postbac Poster Day 30/04/2026

STUDENT SUPERVISION

Google Summer of Code (GSoC) mentor, NRNB

05/2026 - present

- Snehashree Prusty - Direct Integration of PathVisio with WikiPathways GitHub

Google Summer of Code (GSoC) mentor, NRNB

05/2024 - 09/2024

- Aditya Saini - GPML support in Newt
- Tushar Chowdhury - Cytoscape layout plugin to support Cytoscape.js layouts

Co-supervision / Research Mentoring

09/2016 - 07/2023

- Alejandro Adriaque Lozano, MS, MaCSBio, Maastricht University
- Joshua Muller, BS, MaCSBio, Maastricht University
- Osama Zafar, MS, Bilkent University
- Mubashira Zaman, MS, Bilkent University
- Alihan Okka, MS, Bilkent University

TEACHING EXPERIENCE

Practical instructor / lecturer / tutor

01/2023 - 01/2024

Maastricht University, NL

- Practical instructor for Network Biology (Bachelor Biomedical Sciences)
- Proposal writing coach for Systems Biology (Master Systems Biology program)
- Lecturer for Network Biology (Master Systems Biology program)
- Tutor for Network Biology (Master Systems Biology program)

Teaching Assistant

06/2012 - 08/2022

Bilkent University, Turkey

- CS473 - Algorithms I, CS421 - Computer Networks, CS319 - Object-Oriented Software Engineering, CS202 - Fundamental Structures of Computer Science II, CS201 - Fundamental Structures of Computer Science I, CS102 - Algorithms and Programming II, CS101 - Algorithms and Programming I

INTERNSHIPS

Summer Intern

06/2011 - 07/2011

ASELSAN, Turkey

- Developed a database application to be used as an inventory list by using C# and MSSQL

Summer Intern

06/2011 - 07/2011

The Central Bank of the Republic of Turkey, Turkey

- Developed a database application for the purchasing department by using PHP and MySQL

SKILLS

Programming Skills: JavaScript, Java, R, Python, HTML, CSS, Matlab, Cypher, SQL

Libraries and Tools Cytoscape.js, Node.js, React, Git, Docker, jQuery, OpenGL, Unity

Databases Neo4j, MySQL

Operating Systems Linux, Windows, MacOS

AWARDS AND HONORS

Postdoctoral Visiting Fellow Award, NIH Visiting Fellow Program, National Library of Medicine 2024

Full Graduate Scholarship, PhD, Bilkent University 09/2016 - 08/2022

Graduate Scholarship, The Scientific and Technological Research Council of Turkey 09/2012 - 08/2014

Full Graduate Scholarship, MS, Bilkent University 09/2012 - 08/2015

Graduation with High Honor Degree, BS, Computer Engineering, Bilkent University 06/2012

Senior Design Project Innovation Award 05/2012

Full Undergraduate Scholarship, Bilkent University 08/2007 - 06/2012

Achieved 1167th position in University Entrance Exam (among 1.5 million students) 06/2007